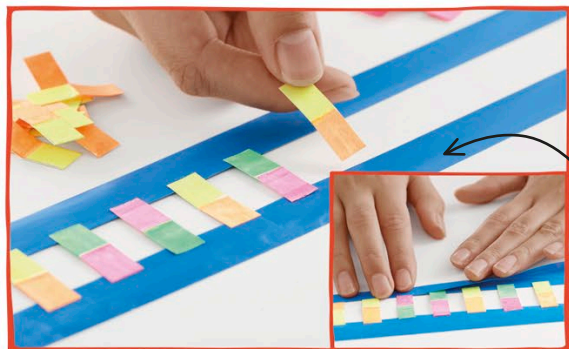




3 Now color your paper rungs on both sides. One half should be one color, the other half another. The colors should always be in pairs—for example, yellow always goes with orange.

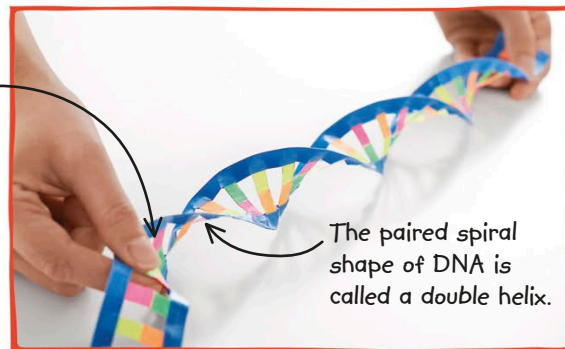


4 Now cut two strips of tape about 28in (70cm) long. Lay the strips face up next to each other, with a gap of $\frac{3}{4}$ in (2cm) between them. Stick down the ends with small pieces of the same tape.



Hold and twist the ladder very carefully.

Everybody's DNA is different, so there is no right or wrong combination.



5 Press the colored rungs, in any order, onto the two lengths of tape, leaving a gap of about $\frac{1}{2}$ in (1cm) between them. When you've used up all the rungs, carefully fold both lengths of sticky tape over them to hold them in place.

6 There's just one more thing to do to make your model perfect: you have to twist the ladder into a spiral shape that's just like DNA itself. Do this very gently to get the twist right, turning the end nearest to you in a counterclockwise direction.

HOW IT WORKS

The bases in DNA—as represented in the model you created—are a code of instructions for how to make proteins. These are large, complex molecules required for the structure, function, and regulation of the body's tissues and organs. For example, the protein keratin makes up your hair and nails. A section of the DNA ladder that carries a recipe for a particular protein is called a gene. Your entire DNA code consists of about 20,000 genes. Together, your full set of genes is called a genome. No one else has exactly the same genome as you—unless you have an identical twin.

REAL WORLD SCIENCE DNA SEQUENCE



Using special equipment, scientists can determine what they call a DNA sequence: the exact arrangement of the bases along the length of a DNA molecule. This means samples can be used to identify people or to spot genes that can cause diseases.